Fairness Preservation under Computational Budgets in Probabilistic Decision Systems

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 9
Shreeya Kamath, 261090830022; Misha, 261090830020
Repository:
`https://github.com/ShreeyaKamath/Fairness-Preservation-and-Computational-Budgets-AFGML`
Experiment-producing commit: `e2350bcd6d58361ddf3e4e7adcdc5d8383f586c2`; submission tag `submission`;
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Abstract

We ask whether a budget-aware fallback limits the fairness loss when exact inference in a nine-variable Bayesian network for pre-contact bank outreach is cut off by an operation budget. On UCI Bank Marketing (test $n = 8{,}238$), fairness-aware fallback selection reproduces exact decisions (AP 0.421, 98.3% agreement) but has a larger Δ_{EOpp} than the global fallback by $+0.120$ $[0.015, 0.226]$; our hypothesis is refuted. Simulation shows group-dependent evidence availability drives fallback disparity. Limits: strong base-rate drift, no customer identifiers, small network.

1 Question, context and contribution

Customers, outreach teams, serving engineers and auditors are affected when a probabilistic scorer runs under a compute limit. $Y = 1$: recorded term-deposit subscription; $D = 1$: selection for *hypothetical* outreach (top 20% of scores) using only pre-contact information (call `duration` excluded).
Research question: at a fixed operation budget, does choosing the fallback by validation worst-group deviation from the exact decision (AP loss ≤ 0.01) yield a smaller Δ_{EOpp} than a global fallback, utility-only selection and post-budget threshold adjustment? It is refuted if any Holm-adjusted paired interval contains 0 or is positive.
Contribution: a deterministic operation-count budget with evidence-dependent trips, a simulation isolating group-dependent evidence availability (with a no-dependence control), and the selection rule.
Closest work: compute reduction causing group disparity in neural inference [1]; our reproduced baseline is group-threshold post-processing [3]. Eight or more papers are compared in `docs/related_work.md`.

2 Data, graphical model and fairness method

Data and estimand: `bank-additional-full.csv` [2] (41,188 contacts, 4,640 positives, SHA-256 recorded), date-ordered, no customer id; positional 60/20/20 split (24,712/8,238/8,238). Protected attributes: age band (< 30, 30–59, ≥ 60) crossed with marital status (audit only; not a model input). Groups with < 30 positives are exploratory. `unknown` values are unobserved evidence; macro and calendar columns are excluded by declared choice.
Graph and inference: a Dirichlet-smoothed (train-only) discrete Bayesian network, no latent variables; A age band, E education, J job, F default, H housing, L loan, P_d, P_r prior-contact bins, O previous outcome:

$$p(\cdot) = p(A)\,p(E|A)\,p(J|A,E)\,p(F|J)\,p(H|A)\,p(L|H)\,p(P_d)\,p(P_r|P_d)\,p(O|P_r,P_d)\,p(Y|A,J,H,L,O,P_d),$$

query $P(Y{=}1 \mid e)$ with e the observed variables. Exact variable elimination (min-fill, min-weight, min-degree) and clique-tree calibration (treewidth 6) were implemented with factor-entry operation counters.
Method and fairness: a query whose exact cost would exceed the budget is answered from a precomputed table (prior, or a variable subset). The proposed rule picks, on validation data, the candidate with the smallest worst-group decision-flip rate relative to exact inference among those within the AP tolerance; routing is group-unaware. Primary criterion Δ_{EOpp} (range of group TPRs; a missed responder is the harm); diagnostics Δ_{DP}, Δ_{EOdds}, flips, log loss, Brier. Primary utility:

average precision (AP), allowed loss 0.01. Age band is a model variable and so is used at prediction time; marital status is not.

3 Evaluation protocol

Baselines: exact inference (reference), global fallback (closest), threshold adjustment after budgeted inference, fair-adjusted exact, logistic regression and gradient-boosted trees on the same variables. Thresholds, tables and selection use train/validation only; the test protocol was frozen beforehand (`docs/protocol.md`). Budgets are operation counts at quantiles of natural validation cost; the primary 75% quantile is 11 operations (258 at 100%); wall-clock is measured separately. Capacity is 0.2 (robustness 0.1, 0.3). Workloads: natural (real `unknown` pattern, confirmatory) and synthetic group-dependent / group-independent masks (exploratory). Five seeds cover masks, networks and GBT; the natural workload is deterministic. Intervals: fixed-model paired moving-block bootstrap over date-ordered test blocks (length 50, 1,000 replicates, 95%); Holm correction over the three confirmatory comparisons, computed on Δ_{EOpp} over the 7 groups with ≥ 30 positives. Ablations: utility-only selection, global fallback, no evidence-based simplification, generous budget. Robustness: elimination orders, network densities, discretisation, capacities, budgets, masks. Correctness: exact methods agree to $\leq 2.2 \times 10^{-16}$ (tolerance 10^{-6}) on 600 evidence patterns and unit tests in `tests/`.

4 Results and interpretation

Method	AP ↑	Δ_{EOpp} ↓	Worst-group flip ↓	Log loss / Brier ↓
Exact (reference)	0.421	0.465	0	0.709 / 0.223
Global fallback (closest baseline)	0.350	0.303	0.303	0.852 / 0.260
Proposed	0.421	0.423	0.038	0.710 / 0.223
Utility-only selection (ablation A1)	0.354	0.717	0.198	0.825 / 0.254
Threshold adj. after budget	0.350	0.317	0.319	0.852 / 0.260

Table note: natural workload, test n = 8,238, 11-operation budget (20.1% of queries fall back), capacity 0.2, one deterministic run. Δ_{EOpp} over 7 supported groups (70–932 positives each); 5 exploratory groups (1–8 positives, n = 2–34) excluded. Paired proposed − comparator on Δ_{EOpp} (negative favours proposed): global $+0.120$ $[0.015, 0.226]$, Holm $p = 0.036$; utility-only -0.294 $[-0.370, -0.210]$, $p = 0.006$; threshold adj. $+0.106$ $[-0.006, 0.251]$, $p = 0.080$.

Main finding: the hypothesis is refuted. Proposed beats utility-only selection, but is worse than the global fallback and indistinguishable from threshold adjustment. The lower Δ_{EOpp} of those two reflects levelling down: AP falls by 0.071 (above the 0.01 tolerance) and 15–18% of decisions change, whereas proposed stays within tolerance (98.3% agreement). Against exact inference its Δ_{EOpp} is lower by 0.042 $[0.006, 0.082]$: no budget-induced degradation, but the 0.465 gap already exists in the exact model. The BN's AP is below logistic regression (0.525).

Mechanism and failure: removing fairness-aware selection (A1) raises the worst-group flip rate from 0.038 to 0.198, as it picks the age-based fallback. Disabling simplification (A3) raises fallbacks to 26.2%, and a generous budget (258 operations) restores exact behaviour. In simulated networks the fallback-rate gap is 0.542 under group-dependent availability versus 0.026 in the control (difference $+0.516$; each width's interval excludes 0), yet the Δ_{EOpp} effect is uncertain at one width ($+0.069$ $[-0.041, 0.179]$). Failure (synthetic dependent masks, exploratory): proposed chose the age-based fallback in 3 of 5 seeds (AP loss 0.046; Δ_{EOpp} $+0.493$ $[0.397, 0.569]$ versus exact), while its group-unaware variant lost 0.003 AP. It also degrades under dense and sparse networks (worst flip 0.206, 0.133).

5 Limitations, ethics and conclusion

Only the natural `unknown` pattern is real; masks and networks are simulations. The positive rate drifts across splits (4.8%, 11.1%, 30.8%), so the validation threshold selects 38.3% of test rows instead of 20%. Rows are not distinct customers, so intervals may be too narrow; five groups are exploratory and wide intervals do not show fairness. Contacted-only data say nothing about uncontacted people, and with nine variables exact inference is cheap, so budgets are relative. **Conclusion:** fairness-aware fallback selection preserves exact decisions at a tight budget but does not give the smallest Δ_{EOpp}; next, re-run under a re-split that equalises base rates and add worst-group TPR to expose levelling down.

Reproducibility and individual contributions

Reproduce: `scripts/run_experiments.py -mode real`, then `scripts/evaluate.py -out results`, then `scripts/make_results.py -out results -mode real`; config: `configs/main.yaml`; Python 3.13, Linux, about 22 minutes, `requirements.txt`; commit `e2350bcd6d58`.
Shreeya Kamath: implementation (inference, baselines, experiments, notebook) and slide alignment; checked the report and slide figures against `results/`. **Misha:** debugging, report writing and slides; re-ran the 93 tests and the exact-agreement and leakage checks on Shreeya's code. Details: `docs/contributions.md`.
Evidence: `docs/requirements_matrix.md`, `results/`.

References

[1] S. Das, M. Romanelli, F. Fioretto. Disparate Impact on Group Accuracy of Linearization for Private Inference. ICML 2024 (PMLR 235:10168–10184). arXiv:2402.03629.

[2] S. Moro, P. Rita, P. Cortez. Bank Marketing, UCI Machine Learning Repository, dataset 222. `https://archive.ics.uci.edu/dataset/222/bank+marketing`; accessed 7 Nov 2026.

[3] M. Hardt, E. Price, N. Srebro. Equality of Opportunity in Supervised Learning. NeurIPS 2016. arXiv:1610.02413.